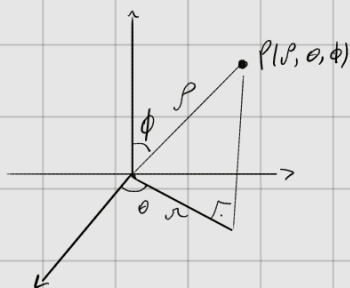


Coordenadas esféricas:



$$\rho(\text{raio}) \rightarrow \rho \geq 0$$

$$\theta(\text{têta}) \rightarrow 0 \leq \theta \leq 2\pi$$

$$\phi(\text{fi}) \rightarrow 0 \leq \phi \leq \pi$$

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

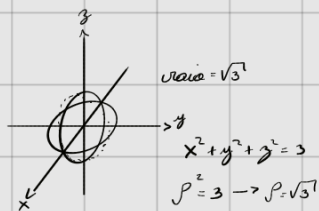
$$z = \rho \cos \phi$$

Jacobiano:

$$\left| \frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} \right| = \rho^2 \sin \phi$$

$$x^2 + y^2 + z^2 = \rho^2$$

Calcular $\iiint dV$ limitada pelo sólido $x^2 + y^2 + z^2 = 3$, na primeira octante



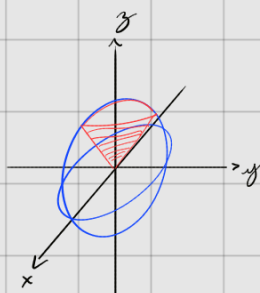
* Sólido primeira octante

$$\begin{cases} 0 \leq \theta \leq \frac{\pi}{2} \\ 0 \leq \phi \leq \frac{\pi}{2} \\ 0 \leq \rho \leq \sqrt{3} \end{cases}$$

$$f(\rho, \theta, \phi) = \rho^2 \sin \phi$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^{\sqrt{3}} \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi &= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \left[\frac{\rho^3}{3} \right]_0^{\sqrt{3}} d\theta \, d\phi = \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin \phi \cdot \sqrt{3} \, d\theta \, d\phi = \sqrt{3} \int_0^{\frac{\pi}{2}} \sin \phi \cdot \theta \Big|_0^{\frac{\pi}{2}} d\phi \\ &= \sqrt{3} \cdot \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin \phi \, d\phi = \frac{\pi \sqrt{3}}{2} \cdot (-\cos \phi) \Big|_0^{\frac{\pi}{2}} = \frac{\pi \sqrt{3}}{2} \cdot (0 - (-1)) = \frac{\pi \sqrt{3}}{2} \end{aligned}$$

Calcular $\iiint e^{\sqrt{x^2+y^2+z^2}}$ limitado por uma esfera no origem e raio 4 e o cone $z = \sqrt{3(x^2+y^2)}$



$$x^2 + y^2 + z^2 = 16 \rightarrow \rho = 4 \quad \rho \geq 0$$

$$z = \sqrt{3(x^2+y^2)} \rightarrow \rho \cos \phi = \sqrt{3(\rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta)} = \sqrt{3(\rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta))} = \sqrt{3\rho^2 \sin^2 \phi} = \rho \sin \phi \sqrt{3} \rightarrow \textcircled{1}$$

$$\textcircled{1} \quad \rho \cos \phi = \rho \sin \phi \sqrt{3} \rightarrow \frac{\cos \phi}{\sin \phi} = \frac{\sqrt{3}}{1} \therefore \phi = \frac{\pi}{6} \quad \text{Jacobiano: } \rho^2 \sin \phi$$

$$\begin{cases} 0 \leq \rho \leq 4 \\ 0 \leq \phi \leq \frac{\pi}{6} \\ 0 \leq \theta \leq 2\pi \end{cases} \quad \iiint e^{\sqrt{x^2+y^2+z^2}} = \int_0^{2\pi} \int_0^{\frac{\pi}{6}} \int_0^4 e^{\rho} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\frac{\pi}{6}} \frac{\rho^3}{3} e^{\rho} \sin \phi \, d\phi \, d\theta = \int_0^{2\pi} \left[-\frac{e^{\rho}}{6} \sin \phi \right]_0^{\frac{\pi}{6}} d\theta = \int_0^{2\pi} -\frac{e^4}{6} \sin \frac{\pi}{6} d\theta = -\frac{e^4}{6} \cdot \frac{1}{2} \cdot 2\pi = -\frac{\pi e^4}{6}$$

$$f(x, y, z) = e^{\sqrt{x^2+y^2+z^2}} = e^{\rho} = e^{\rho^3}$$

$$\int_0^4 e^{\rho^3} \rho^2 \sin \phi \, d\rho = \int_0^4 e^u \frac{du}{3\rho^2} = \frac{1}{3} \sin \phi \int_0^4 e^u du = \frac{\sin \phi}{3} e^u \Big|_0^4 = \frac{\sin \phi}{3} (e^{64} - 1)$$

$$\int_0^{\frac{\pi}{6}} \frac{\sin \phi}{3} e^{64} d\phi = \frac{1}{3} e^{64} (-\cos \phi) \Big|_0^{\frac{\pi}{6}} = -\frac{e^{64}}{3} \left(\cos \frac{\pi}{6} - \cos 0 \right) = -\frac{e^{64}}{3} \left(\frac{\sqrt{3}}{2} - 1 \right) = -\frac{e^{64}}{3} \left(\frac{\sqrt{3} - 2}{2} \right)$$